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Final Project Reflection Essay

Overall, most of the steps taken in the midterm project with manual curation were relatively simple to implement in a more automatic manner. After an initial learning curve of figuring out how to use a config file, listing out the columns to be included as well as their types was easy to do compared to manually picking columns in part one. Additionally, it was easy to go through all selected columns to check their types and ranges rather than manually converting columns. Removing null values was also easy to implement. The hardest part of converting everything to occur automatically was that there was less judgement involved, and the rules had to be general enough that they would work for more cases. In the manual curation, I looked at columns individually and compared between the csv file to see what good data should look like. However, in an automatic pipeline, you cannot make as many specific decisions, you have to make overall decisions. For example, in the manual curation, I could look at each column and determine if it was a good idea to remove the null values for that column. However, in a pipeline, it must be known in advance which columns to remove null values from and which to not remove null values from.

The system handles the data itself well reliably. Once the config file is created and there are no errors in reading it in, then the actual curation steps, such as type normalization and removal of null values, happens smoothly. However, differences in datasets and config files is something that still needs human review. For example, one type of encoding when reading the csv files worked for one of the datasets but not the other, so I had to change it to a type that works for both. This type of encoding might not work for all csv files though, so that likely needs human review. Additionally, the config file structure is relatively specific, and if certain variables

aren't of certain types, for example, lists, errors may occur. Thus, users must ensure that the config file is similar in structure to the original config file. Users must also be careful what strings they put into the config file that count as null values, as an empty string deleted all rows for one dataset but not for another. So, humans should review the curation output to ensure that the number of rows removed and what they were removed for seems reasonable.

In terms of generalization to the second dataset, the workflow generalized relatively well. The second dataset was a lot cleaner than the first dataset, so the pipeline does not make many changes to the data since it is already good quality. I was hoping to find some messy data for the second dataset, but since many surveys are multiple choice, it seems a lot of them don't have missing or messy values. As I stated previously, the type of encoding when reading in the csv file did not work for the second dataset, so that is something that I had to change. Additionally, in the second dataset, it had some questions that are follow-up questions to previously asked questions in the survey. Thus, these columns are allowed to have null responses because only people who answered yes to a previous question would see them. So, if these null rows were eliminated, it would be eliminating all of the people who answered no to a certain question, introducing bias. Thus, in the config file, I added a place to add a list of columns which are allowed to have null values. This way, if there are questions in the survey which may not be asked to everyone, they can be excluded from the null filtering that occurs as part of the curation. The hardest part of using the second dataset was having to go through again and list the columns and types to be included, which is a limitation of the current workflow. For a huge dataset with many columns, it would be very difficult to go through all of those columns, list them out, figure out what type they should be, and include a range of values. Overall though, the workflow generalized well to the second data set.

This pipeline makes a few different assumptions about clean data. Rather than normalizing null values, it removes them entirely (unless the columns they are part of are specifically included as allowed to be null in the config file). Additionally, it only normalizes the type of the column and removes out of range values, since that is what the datasets I worked with needed to be cleaned. Since the datasets I worked with were mostly clean, there was no normalization of the actual data itself within columns. This may be an issue for messier datasets, such as if leading zeros need to be removed or something similar. It is also assumed that columns will easily be able to be converted to another type, as that is what worked for the datasets I used. This could lead to errors though if data that cannot be converted exists in a column that needs to be converted. Finally, the system assumes that the number of rows removed is the most important metric to be reported, and that may be untrue depending on the goals of the curator. The curator may want to know exactly which rows were removed, which is currently not what the system does; it says why rows were removed, but does not output those removed rows.

There are a few different risks present in my pipeline. As stated previously, because null values are removed rather than filled with a value, that could lead to bias or erasure if there is a systematic difference in the type of respondents to have null values in their responses compared to those who do not. For example, if for some reason women were more likely to have null values in their survey responses, then this curation pipeline's output would be biased toward men because more women's responses would have been removed. However, I believe that removal is a better solution than putting the average value in for a null value because that may lead to inaccurate results/conclusions. An additional risk is that if the curator forgets to include a column in the config file, then that column will not be included in any of the output/further uses of the curated data. This mistake may occur especially in very large datasets, and this could lead to

inaccurate or incomplete results if not all of the data is included. Finally, having an automatic pipeline could result in people feeling as if they do not have to check the curation work that is done. However, this is not the case, and there should still be human oversight/checking of the steps that are taken by the pipeline; that's what the curation log is for.

Multiple improvements would have to be made before this system could be used professionally. Firstly, a better way of stating which columns should be used could be implemented. As I have stated previously, having to list the columns to be included introduces a lot of human error. Additionally, having the pipeline automatically know what type a certain column should be without it being included in the config file would be helpful and save time as well. If a functionality could be introduced to read through a codebook provided for a dataset and create the config file from that, which could then be checked by a human, it would save a lot of time and possibly also reduce errors in the creation of config files since humans would just have to check it rather than create the whole thing themselves.

Moreover, the system would have to be customized further for the specific data that the company wants to use it on. The company would have a better idea of what problems usually occur in the types of data that they use, so steps could be added to the curation pipeline to address those specific issues. For example, if a company uses data where sometimes they want to replace a null value with the average for that column, that functionality could be added to the function which removes null values. Or, if a company knows that they often have a lot of messy strings, functionality could be added to normalize those string values to make them more readable. Right now, the system is more generalized for survey data overall, but it could be made much more detailed for a specific company's purposes.

All in all, the current pipeline works well overall but has some limitations and areas in which it could be improved. It does a good job at removing null values, doing type conversion, removing out of range values, and creating descriptive curation output. However, it could be improved for use in more specific domains, and it could be further tested on more datasets to ensure that it is functional in as many cases as possible.